Haibo Liu

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School of Engineering, Westlake University, Hangzhou 310030, P.R. China.

Education

Tsinghua University Beijing, China (2019/8-2025/6)
Ph.D. in Mathematics. Advised by Prof. Junqing Chen

Lanzhou University Lanzhou, China (2015/8-2019/6)
B.S. in Mathematics and Applied Mathematics

Georgia State University Atlanta, USA (2017/6-2017/9)
Exchange in the Department of Mathematics and Statistics

Employment

Research Assistant, Westlake University Hangzhou, China (2025/7-now)

Research Interests

My research interests cover Graphics, 3D Computer Vision and Inverse Problems. More specifically:

- 3D Generation
- Inverse Problems (Shape Optimization)

Publications

- DiLO: Decoupling Generative Priors and Neural Operators via Diffusion Latent Optimization for Inverse Problems, Preprint, (2026).
- Generative Prior-Guided Neural Interface Reconstruction for 3D Electrical Impedance Tomography, Journal of Computational Physics, (2026).
- KeyframeFace: From Text to Expressive Facial Keyframes, Preprint, (2025).
- Solving inverse obstacle scattering problem with latent surface representations, Inverse Problems, (2024).
- An Alternating Direction Method of Multipliers for Inverse Lithography Problem, Numerical Mathematics: Theory, Methods and Applications, 16 (2023).

Awards and Honors

- Outstanding Doctoral Dissertation of Tsinghua University (2025)
- National Endeavor Scholarship, Lanzhou University of China (2016)
- The 9th National College Math Competition National Second Prize, China (2018)

Scientific Events

- Co-organizer: "0-1 Seminar", Tsinghua University (July 2022)

Teaching at Tsinghua University

- Associate teaching experience: Probability theory and mathematical statistics (Autumn 2020)
- Associate teaching experience: Calculus (Spring 2020)
- Associate teaching experience: High numerical analysis (Autumn 2019)

Seminar and Conference Talks

- The 631st Doctoral Student Forum of Tsinghua University (April 2024).